

"Statistical Foundation for Data Science"

Unit 1: Introduction to Statistics

1. Define statistics. Explain its scope and importance in data science.
2. What are the types of data (qualitative vs quantitative)?
3. Define population and sample with examples.
4. Explain the types of statistical variables (discrete and continuous).
5. Distinguish between descriptive and inferential statistics.

Unit 2: Measures of Central Tendency

1. Define mean, median, and mode. Explain with examples.
2. Calculate the arithmetic mean for grouped and ungrouped data.
3. Explain merits and demerits of different averages.
4. How to find median and mode using formulas and graphical method?
5. Distinguish between mean, median, and mode.

Unit 3: Measures of Dispersion

1. Define range, variance, and standard deviation.
2. Explain the computation of standard deviation for grouped data.
3. What is mean deviation? How is it calculated?
4. Explain the coefficient of variation and its uses.
5. Distinguish between absolute and relative measures of dispersion.

Unit 4: Correlation and Regression

1. Define correlation. Explain types of correlation with examples.
2. Explain Karl Pearson's correlation coefficient with formula.
3. What is Spearman's Rank Correlation? Give steps to calculate.
4. Define regression. Write the regression equation of Y on X.
5. Distinguish between correlation and regression.

Unit 5: Probability and Distributions

1. Define probability and explain its basic rules.
2. State and explain addition and multiplication theorems of probability.
3. What is conditional probability?
4. Explain binomial distribution and its properties.
5. Define normal distribution and its significance in data science.